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Enhancing Ransomware Detection - A Comparative Review of XGBoost, Random Forest, and Neural Network Approaches

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***Abstract*— Ransomware detection has now become an essential challenge in cybersecurity as attackers use using smart and tricky methods to get interrupts our simple defences. This review discusses how the work of leading machine learning (ML) models like, XGBoost, Random Forest (RF), and Artificial Neural Networks (ANNs), improves ransomware detection systems. The most prominent reason why XGBoost has been selected as the model is because of its high accuracy rate, and random forests are considered due to their appreciation for model stability, which is why they have been used within studies for efficient base classifiers for most of them. ANNs also offer a strong nonlinear approach when handling ransomware's complex data patterns. In this paper, we evaluate some of these approaches- hybrids that include XGBoost and RF algorithms as well as advanced ANNs configurations for their very distinct strengths and weaknesses in the sense of ransomware detection. In synthesizing the results, The review explains how applicable and comparatively advantageous these approaches are at detecting ransomware. The study ultimately seeks to guide future implementations by summarizing key model features, evaluation metrics, and adaptability to evolving ransomware behaviours, there by contributing to enhanced cybersecurity frameworks.**

Keywords*— *Ransomware Detection, Machine Learning (ML) Models, XGBoost, Random Forest (RF), Artificial Neural Networks (ANNs), Cybersecurity Frameworks.

I. INTRODUCTION

Ransomware attacks are a strong threat to cybersecurity for people, businesses, and critical infrastructures around the world. With their increasingly sophisticated evasion techniques, the need for real-time strategies for ransomware-detection increases. Machine Learning and Deep Learning have generated a lot of interest in terms of their applications as technologies that learn behavioral patterns from ransomware for detection. Most algorithms widely used today are XGBoost, RF, and ANNs .Research focused on intelligent models based on artificial intelligence for ransomware detection. According to one study, it also provides an overview of existing ML techniques, which will greatly benefit researchers in the area [1]. On the contrary, feature selection using stacked autoencoders improves the ransomware signatures detected, better than models such as XGBoost [2]. There was a better network detection performance when Bayesian Neural Networks (BNNs) and Slab Bayesian Neural Networks (SBNNs) are paired with neural networks and with statistical paradigms [3]. Import Monitoring of Portable Executable (PE) with especially Logistic Regression has been pretty well proven useful in distinguishing the ransomware from benign programs [4]. Moreover, Graph Neural Networks (GNNs), such as Graph Convolutional Networks (GCNs), have outperformed Graph Attention Networks (GATs) for ransomware prevention [5]

Some advanced techniques about deep learning, for example, recurrent neural networks (RNNs), provide more efficient patterns classification to enhance early ransomware detection [6]. Particle swarm optimization combined with ensemble machine learning techniques have also decreased false positive rates while improving detection rates [7]. Unlike previous proposals, ANNs using Power Iteration technique with PE headers have been shown to be precise in the detection of ransomware families [8]. Weighted generative adversarial networks (GANs) were able to create synthetic samples for addressing zero-day attacks and enhancement in detection accuracy [12]. Hybrid approaches, such as CNN and Random Forest, combining models that analyze networks behavior showed high accuracy [23]. K-means clustering together with SVMs have incorporated attack early detection within enterprise environments [24]. Explainable AI (`XAI') transforms the prediction transcript reliability to inspire trust in the security personnel toward the AI judgment [26].

Neural network advances have produced technologies like Long Short-Term Memory (LSTM) networks to analyze file system behavior, and Deep Belief Networks (DBNs) for multi-level feature extraction, which have promising prospects for addressing future evolutions in ransomware attack processes [28]. Op-code analysis together with naive Bayesian techniques strengthened the protection against these attacks, while predictable patterns of bitcoin transactions were efficiently traced back to ransomware payments as evidenced by neural network research [10].

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The review therefore captures the benefits of ML algorithms, possible mechanisms for improvement, and future feasibility in terms of real-time protection from ransomware in modern cybersecurity systems.

II. LITERATURE REVIEW

Almost all the frameworks of ransomware detection are based on machine learning (ML), wherein the dynamic characteristics of ransomware are analysed [1]. In feature selection by a stacked autoencoder, high classification accuracy was attained, which surpasses XGBoost [2]. By using the radial spike and slab models, the performance of Bayesian Neural Networks classified sparse data improved when used in association with Bayesian Statistics and neural networks [3]. Network traffic classification with Decision Tree (DT) and XGBoost achieved more than 99% accuracy in hostile software detection [4]. Logistic Regression is highly accurate for the analysis of imports from the Portable Executable (PE) format, distinguishing between ransomware and clean files [5]. Graph Convolutional Networks (GCNs) performed better compared to Graph Attention Networks (GATs) regarding detection for cyber threats [6].

Deep learning like Recurrent Neural Networks (RNNs) is applied in the early discrimination of ransomware types. However, there are limitations on pace in actual practice [7]. Ransomware detection was established through analysis of behavioural file system activities by Event Tracing for Windows (ETW) [8]. Power iteration and ANNs improved their performance of detection through extracting information from PE headers as [9]. Neural networks were employed to analyse bitcoin transactions and showed very promising results [10], while MLP confirmed their usefulness in detecting emerging families of ransomware [11]. Zero-day attacks are addressed through synthetic sample generation referring to accuracy improvements in detection made by weighted GANs. False positives were reduced by Adaptive Graph Neural Networks, thereby enhancing their efficiency [13]. Superior classification accuracy was demonstrated by Random Forest (RF) models, especially in ransomware versus benign file analysis [14].

The upgraded models had a random forest approach, improving the analysis of opcode sequences hence increasing the accuracy of Ransomware identification [17]. By using a behavioural analysis, effective classification of ransomware was made using an SVM and PCA model [18]. A data gathering strategy with two tiers for network and file traffic as well as novel clustering methods was able to really improve detection of ransomware [19]. Isolation Forest and RF models were integrated into infrastructure protection systems followed by ISO standards [20]. Text mining and ML leveraged TF-IDF for processing API calls and network interaction [21]. Analytical bitcoin addresses in ransom payments accesses the malicious activities in the blockchain [22]. The combination of CNN with Random Forest models improved the accuracy of detection by networks [23].

K-means clustering and SVM have proven to be effective for the early detection of corporate environments [24]. Decision Tree and Gradient Boosting frameworks monitored attack patterns in real-time with improved precision [25]. Explainable AI (XAI) provided better reliability and trust in ransomware predictions [26]. RNNs analysed polymorphic ransomware logs for strong profiling [27], while LSTM networks identified advanced ransomware attacks by learning file system behaviours [28]. Naive Bayes and opcode analysis effectively addressed the zero-day ransomware threats [29]. Random Forests enhanced detection efficacy while also minimizing latency through dynamic feature extraction [30]. Deep Belief Networks (DBNs) were able to extract features across multiple layers, demonstrating their power and precision in handling new strains of ransomware [31].

III. METHODOLOGY

The methodology is focused on the detailed comparative study of three prominent machine learning techniques employed in ransomware detection in the domain- XGBoost, Random Forest, and ANNs, which have been effective in overcoming distinct challenges posed by the evolving complexity and behavior of ransomware.

A. XGBoost (Extreme Gradient Boosting)

XGBoost is an popular of a machine learning algorithm employing the gradient boosting framework. It is built with emphasis on the computational efficiency, speed and performance with regards to prediction problems. XGBoost employs an ensemble of decision trees as base learners and improves the predictions continuously by adjusting the predictions of the previously built models in the stack. This method has incorporated advanced concepts of optimization such as regularization, coordination of cores for processing, and optimization of memory usage which makes it suitable for both classification and regression analysis. This method is mainly applicable to structured or tabular data and very effective in producing results in modelling contest where machine learning algorithms are put into competition.

XGBoost is developed via an ensemble of decision trees deployed in a sequential manner whereby each succeeding tree complement corrects the errors made in its predecessors. This is where the focus is progressively turned onto the residuals i.e. the variations between the true values as well as predicted values of things, and hence the predictions are sharpened in order to increase accuracy. The central components of XGBoost consist of gradient boosting, which is the core idea of combining a large number of weak learners at the expense of improving the loss function, and regularization techniques such as L1 Lasso and L2 Ridge to mitigate overfitting. Also, the algorithm employs a technique whereby pruning of trees takes place to stop the training after certain rounds, thereby ensuring that the models are neither complex nor hard to explain. To perform well on large datasets, XGBoost employs a weighted quantile sketch algorithm for precise data partitioning together with decomposition for computation acceleration, especially on high-performance systems. In addition, it handles missing values by them being directed to the appropriate child while the splitting of trees is taking place hence no complex preprocessing is required. These advantages lead to the conclusion that XGBoost is robust and versatile when it comes to the issues of data..

The **objective function** for optimization:

$$L(\theta) = \sum_{i=0}^{n} L(y_i, \widehat{y_i}) + \sum_{k=1}^{k} \Omega(f_k) \quad (1)$$

- . $L(y_i, \widehat{y_i})$ is the loss function (usually squared error for regression or log loss for classification)
- $\Omega(f_k)$ is the regularization term, which penalizes the complexity of each tree f_k (the number of splits).
- $\widehat{y_i}$ is the predicted value.
- y_i is the true value for the I th sample.
- θ represents the model parameters.

B. Random Forest Algorithm

A Random Forest (RF)algorithm is a multimodal ML ensemble technique comprised primarily of multiple decision trees. For each individual tree, a random subset of features and data points are used to build the tree which introduces variety and decreases overfitting. During prediction, instead of giving out the result of individual trees, Random Forest combines their result (by averaging in case of regression or voting in case of classification) to present a final output. This combination also strengthens the model and improves its performance on new datasets.

Random Forest offers significant advantages, especially when it comes to ransomware detection. To begin with, it is resistant to overfitting. Because Random Forest combines the outcomes of several decision trees built using different sections of the training data, it minimizes the overfitting effect, which a single decision tree may be prone to. Hence, it is better at predicting unseen variants of the ransomware on which it was not trained. Second, Random Forest is suitable for large volumes of data. It could be processed a large amount of data through the random selection of functional units or sub-features simultaneously, which gives it an advantage over other approaches for commonly encountered data-based problems in cybersecurity. Last but not least, it has good interpretability and high stability. The fact that Random Forest is an ensemble algorithm does not deprive it of transparency because the individual trees are rather simple and easy to understand. These characteristics make RF is an one of the powerful methods of ransomware as well as other malware detection.
Random Forest Equation:

$$f(x) = \frac{1}{T} \sum_{t=1}^{T} f_t(x) \quad (2)$$

- $f(x)$: final prediction of Random Forest model.
- $f_t(x)$: prediction of the t-th tree.
- T : total number of trees in the RF algorithms generate

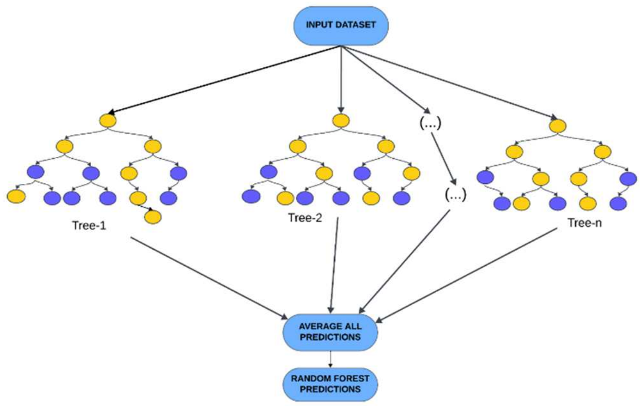

Fig.1 . Implementation of Random Forest Algorithm

This diagram manifests the implementation of the Random Forest algorithm. It describes how an input dataset builds multiple decision trees (Tree-1, Tree-2, ..., Tree-n). Each tree generates a vote prediction and averages the votes to get the overall vote, which is the Random Forest prediction.[30]

C. Artificial Neural Networks:

The use of artificial neural networks for ransomware detection comes through the analysis of structured and unstructured data. Each connection processes input values and signals output to the next layer, refining the future predictions through backpropagation. They are best known for identifying numerous complex records in large-scale datasets such as user activity records, network traffic, and anomaly logs. The models can also be trained on new data to work against the changing aspects of the threats inside the ransomware and on-the-spot threat detection at a very low turnaround time.

Evolved forms of ANNs, such as RNNs for sequential information and CNNs for matrix-type data, are designed for such analysis. Artificial neural networks have employed both the Portable Executable (PE) header and network activity for locating ransomware. The power iteration methods improve the feature extraction accuracy used in malware classification. For example, an unnamed ANN was integrated to monitor Bitcoin transactions associated with ransomware payments. These ANN models essentially capture abnormalities from the windows event logs and scale exceedingly well in enterprise scenarios.

Most of the sophisticated architectures of ANNs manage huge ransomware datasets in real-time with high detection rates. Different architectures of ANNs are used for different purposes. Static analysis is done with the help of Feedforward Neural Networks, for example, for malicious characteristics detection in PE file headers. RNNs, including their advanced version LSTMs, analyze time-sequenced data, which includes system logs or even user behavior, to detect prolonged interactions with ransomware. Traffic patterns, images, biomarkers, and even defects in using traffic patterns in describing a malware signature are processed with the help of CNNs. Auto-encoders are used for anomaly detection and possible dimension reduction for changes in high-dimensional patterns. Usually, these types of ANNs are combined to build hybrid systems for better detection accuracy and versatility.

IV. ML-MODELS

A. Dataset Used

The ransomware detection dataset which was gotten from Kaggle holds a total of 62,485 samples out of which 27,118 are non-malicious files and 35,367 are malicious files. This dataset contains 18 features that encompass both the format and the behavioral aspects of these files. Important features include DebugSize, this is the size of the debug directory, MajorImageVersion, this depicts version of an file, and IatRVA, which is the import address table virtual address. The features that have been provided are meant to show alterations in the system as a result of ransomware attacks which in turn make the dataset relevant for training and testing detection systems [31].

Ransomware's ecologies include ExportRVA and BitcoinAddresses which help in the analysis of ransomware activities given that the two are highly interlinked in most cases with encryptions and ransoms. There is a good ratio of benign samples to malicious samples in the dataset, thus no such bias for the resulting model. Heatmap diagrams of feature correlation enrich the understanding, as they show strong correlations between features that are nursing the engineering and selection. In the heat map, diagonals with bright colors mean self-correlation while variation in the colors of the other region denotes correlation between variable.

TABLE I. DATA SETS CLASSIFICATIONS

S.no	Column	Non-Null Count	D-type
1	DebugSize	62485	int64
2	DebugRVA	62485	int64
3	MajorImageVersion	62485	int64
4	Export-RVA	62485	int64
5	latVRA	62485	int64
6	BitcoinAddresses	62485	int64
7	Benign	62485	int64

This dataset is practical as it reflects the actual scenarios under which ransomware attacks are applicable allowing machine learning models to understand complex and progressive attacks. Random Forest, XGBoost, and Artificial Neural Networks are some of the sophisticated algorithms which use these parameters in ensuring high levels of accuracy, precision and efficiency in preventing ransomware which has made this core in the field houses of science and even application of cybersecurity [31].

B. COMPARISION OF ML-MODELS

A comparative study on the application of various machine learning (ML) models for the detection of ransomware shows that Random Forest has the highest accuracy (99.70%) among other parameters evaluated: precision (0.9976), recall (0.9956), and f1 score (0.9966). XGBoost is the second most accurate after Random Forest with 99.61% accuracy and good rank for feature importance thereby making it possible to detect evolving ransomware in dynamic environments in contrast to the static detection. Adaptable Neural Networks (ANNs) allow for fine-tuning of at 98.50% accuracy but have proven to be useful against evolving threats in ransomware aversion strategies, however still outperformed by the tree-based models as they tend to overfit with complicated data distributions. Gradient Boosting earns similar turbans as XGBoost but slightly lowers in the accuracy scale pegged at 98.90%. Support Vector Machines (SVM) and Decision Trees bolster SSD results (97.80% and 96.10% accuracy) but fall short for online detection because of excessive computational time or the risk of overfitting. Naive Bayes, K-Nearest Neighbors (KNN), and Logistic Regression show even poorer results (94.20%, 95.30%, and 93.80% accurate respectively) and these models are generally more appropriate to small or less complicated datasets. These findings are also consistent with earlier studies for example those which combine Random Forest and XGBoost approaches in the detection of ransomware based on analyzing their behavior and the structure of Portable Executable (PE) files.

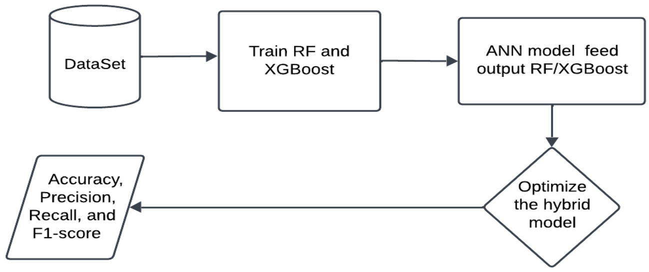

Fig.2 . Process Involves in Comparison of ML Models

The flow chart is represent a hybrid model for ransomware detection. The input is a dataset from which it trains Random Forest and XGBoost models.The hybrid model is optimized to improve performance and, at last, its outcomes are evaluated with metrics such as accuracy, precision, recall, and F1-score.

$$A = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

$$P = \frac{TP}{TP + FP} \tag{4}$$

$$R = \frac{TP}{TP + FN} \tag{5}$$

$$F_1 = 2 * \frac{P * R}{P + R} \tag{6}$$

- True Positives (TP) : Correctly predictions of ransomware.
- True Negatives(TN): Correctly predictions of non-ransomware.
- False Positives (FP) : Incorrectly predictions of ransomware when it is not.
- False Negatives(FN): It fails to detect ransomware.

TABLE II. ML MODELS PERFORMANCE METRICS AND ANALYSIS

Algorithm	Accuracy	Precision	Recall	F1 Score
XGBoost	99.61%	0.9974	0.9938	0.9956
Random Forest	99.70%	0.9976	0.9956	0.9966
ANN	98.50%	0.985	0.983	0.984
Naive Bayes	94.20%	0.943	0.94	0.941
Support Vector Machine (SVM)	97.80%	0.975	0.972	0.973
K-Nearest Neighbours (KNN)	95.30%	0.953	0.95	0.951
Decision Tree	96.10%	0.961	0.959	0.96
Logistic Regression	93.80%	0.938	0.935	0.936
Gradient Boosting	98.90%	0.989	0.987	0.988

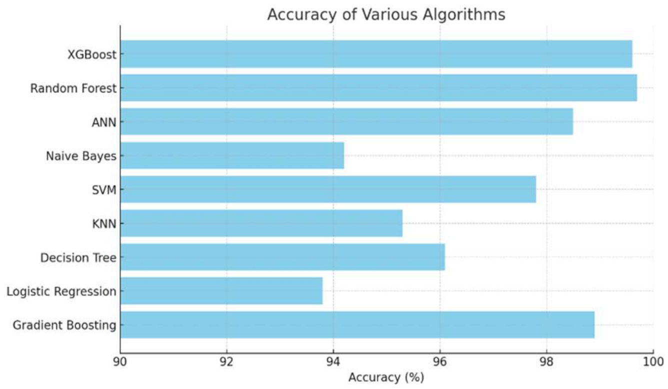

Fig.3 . Analysis Chart for ML Models

This bar chart compares the accuracy of several machine algorithms. Accuracies are shown on the x-axis in percentages, while the top list of algorithms up for comparisons contains XGBoost, Random Forest, ANN, Naive Bayes, SVM, KNN, Decision Tree, Logistic Regression, and Gradient Boosting in y axes. XGBoost and Random Forest show the highest accuracy, soon followed by Gradient Boosting and ANN. Logistic Regression and Naive Bayes, on the other end of the comparison, show the least accuracy compared with these other algorithms. This chart gives a good representation of these algorithms' performances and will be good enough to select appropriate models for individualized research problems.

V. RESULTS AND DISCUSSION

The Comparison of three classifiers have been represented in the performance evaluation section above, which include XGBoost, Random Forest, and Artificial Neural Networks. Random Forest is the highest accuracy of 99.70% with precision, recall, and F1 score of 0.9976, 0.9956, and 0.9966 respectively. This shows that Random Forest is efficient and not only at detecting ransomware, but also controlling false positives and false negatives as a consequence can be used efficiently for ransomware detection.

TABLE III. COMPARISON OF XGBOOST ,RANDOM FOREST, ANN

Algorithm	Accuracy(A)	Precision(P)	Recall(R)	F1 Score (F_1)
XGBoost	99.61%	0.9974	0.9938	0.9956
Random Forest(RF)	99.70%	0.9976	0.9956	0.9966
ANNs	98.50%	0.985	0.983	0.984

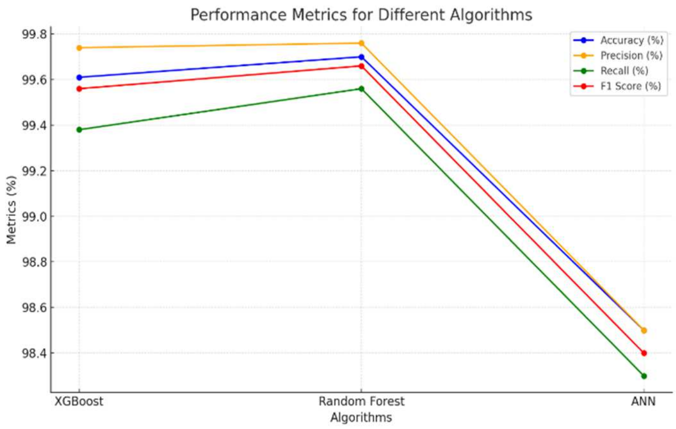

Fig. 4. Performance Metrics of Algorithms

It describes how an input dataset builds multiple decision XGBoost have an accuracy rate of 99.61% with precision and recall values of 0.9974 and 0.9938 respectively yielding an F1 score of 0.9956. Its strong performance indicates its effectiveness when dealing with difficult datasets and that is due to its gradient boosting in which the concept of bias and variance is being reduced as very effective.

While ANN models have underperformed particularly on the accuracy rate when compared with the ensemble models, they managed to achieve an accuracy of 98.50% with precision of 0.985, recall of 0.983 and F1 score of 0.984. On the other hand, even though these approaches are comparatively weaker than the ensemble-based ones, ANNs are superior in capturing non-linearity and learning more complex features from the data.

Based on the above results, Ensemble methods, especially Random Forest and XGBoost, provide outperformance concerning some accesses on ransomware attack detection. Random Forest, based on a tree-bagging approach and super high recall rates (0.9956), proves to be an efficient data classifier with clear results, making it very sensitive and useful for classifying ransomware. However, XGBoost might prove its mettle due to its gradient boost; recall rates, however, fall below those of Random Forest. Also, ANN has hopes of addressing the existing gaps, that is, in the complex and unstructured data present for which future enhancements should include network configuration and feature engineering for improved performance measurement in terms of accuracy and recall. Selecting the right kind of algorithm is vital for efficient ransomware detection, and Random Forest is the best in this area.

VI. CONCLUSION

This review aimed to discuss the effectiveness of XGBoost, RF, and ANNs for the detection of ransomware. In comparison to the other models tested, the Random Forest model was found to be the most accurate, achieving an accuracy of 99.70% with better precision and recall values. XGBoost came in a close second with an accuracy of 99.61% while ANNs proved effective at non-linear pattern recognition with an accuracy of 98.50%. All models have the weaknesses and strengths, pointing to the possibility of each of them being used in solving the problem of ransomware detection.

for Future Research to improve more effective ransomware detection, there is a need to design a hybrid model like the Random Forest, XGBoost as well as ANNs. An ensemble of this sort, would combine the strength of random forest accuracy with the power of optimization from XGBoost and learning from ANNs. This hybrid model is capable of obtaining higher accuracy, precision and recall creating a powerful and flexible system for

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